Experiment 4 Siddhant Bajaj

Aim: To create an account using MetaMask and perform a cryptocurrency transfer from one account to another using a blockchain network.

Objectives

1. To understand the working of the MetaMask cryptocurrency wallet.
2. To create and configure MetaMask accounts.
3. To understand public wallet addresses and cryptocurrency transactions.
4. To transfer cryptocurrency between two accounts.
5. To verify the transaction using a blockchain explorer.

Theory

MetaMask is a cryptocurrency wallet that allows users to create and manage blockchain accounts and interact with decentralized applications. It supports Ethereum and compatible blockchain networks.

Every Ethereum account has a unique public address that is used to receive cryptocurrency. The account is controlled by cryptographic keys, which are used to authorize transactions.

When cryptocurrency is transferred from one account to another, the sender specifies the receiver's address and the amount to be transferred. The transaction is digitally signed and submitted to the blockchain network for processing. The network validates the transaction and records it on the blockchain.

Each transaction has a unique transaction hash that can be used to check its status and details using a blockchain explorer.

For this experiment, the Sepolia test network can be used to transfer test ETH without spending real cryptocurrency. A small transaction fee, known as a gas fee, is required to process the transaction.

Requirements

- MetaMask cryptocurrency wallet
- Google Chrome, Microsoft Edge, or another supported browser
- Internet connection
- Sepolia test network
- Sepolia test ETH
- Blockchain explorer for transaction verification

Procedure

1. Open the official MetaMask website and install the browser extension.
2. Open MetaMask and select the option to create a new wallet.
3. Set up the wallet and securely back up the Secret Recovery Phrase.
4. Open the account interface and verify that the wallet has been created.
5. Enable and select the Sepolia test network.
6. Obtain test ETH from a trusted Sepolia faucet, if required.
7. Create a second account in MetaMask using the account selector.
8. Copy the public address of the second account.
9. Switch to the first account and click the **Send** option.
10. Enter the public address of the second account and specify the amount of test ETH to transfer.
11. Review the recipient address, transfer amount, network, and estimated transaction fee.
12. Confirm the transaction and wait for it to be processed.
13. Copy the transaction hash from the transaction details.
14. Open the Sepolia blockchain explorer and search for the transaction hash.
15. Verify the transaction status and check the updated balance of the receiving account.

Sample Calculation

The final balance of the sender is calculated as:

Final Sender Balance = Initial Balance − Transfer Amount − Transaction Fee

The final balance of the receiver is calculated as:

Final Receiver Balance = Initial Balance + Amount Received

For example, if the sender initially has 0.05 ETH, transfers 0.001 ETH, and pays a transaction fee of 0.00002 ETH:

Final Sender Balance = 0.05 − 0.001 − 0.00002

Final Sender Balance = **0.04898 ETH**

If the receiver initially has 0 ETH:

Final Receiver Balance = 0 + 0.001

Final Receiver Balance = **0.001 ETH**

Result

A MetaMask wallet was created, and two accounts were configured for the experiment. Cryptocurrency was transferred from the sender account to the receiver account using the selected blockchain network. The transaction was verified using its transaction hash on a blockchain explorer.

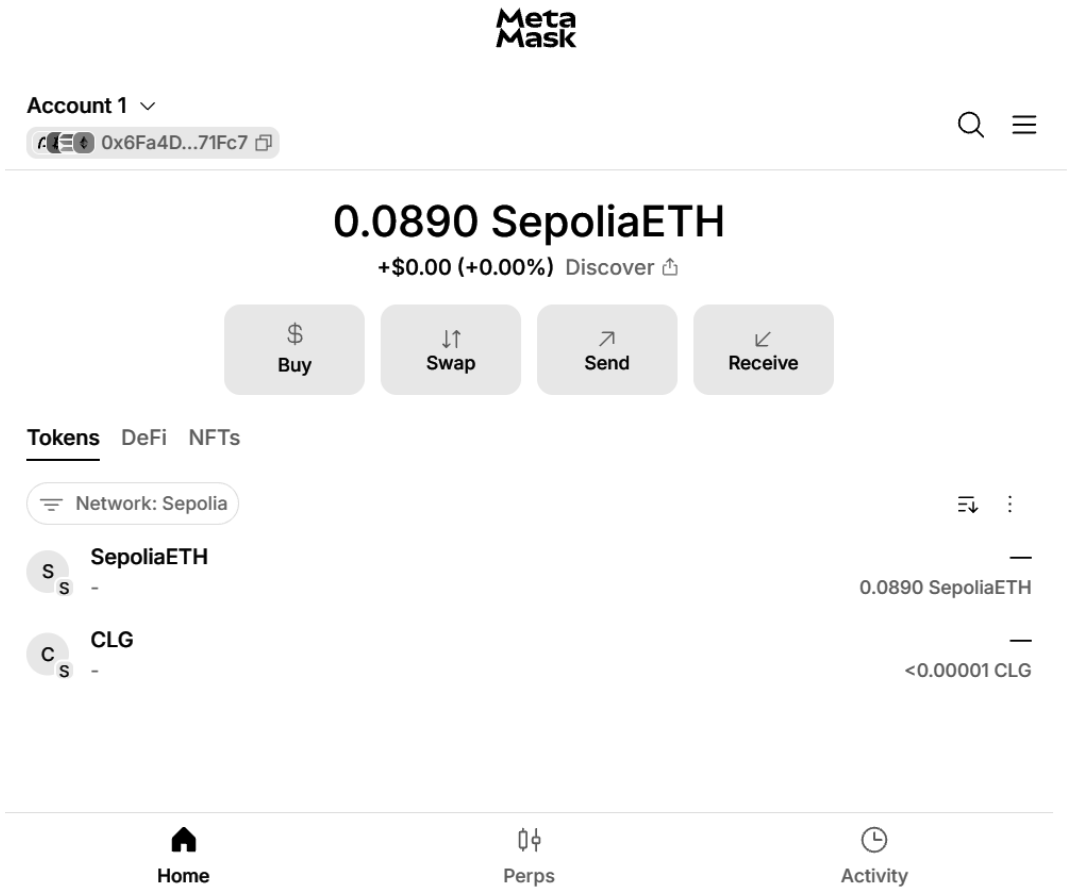
Conclusion

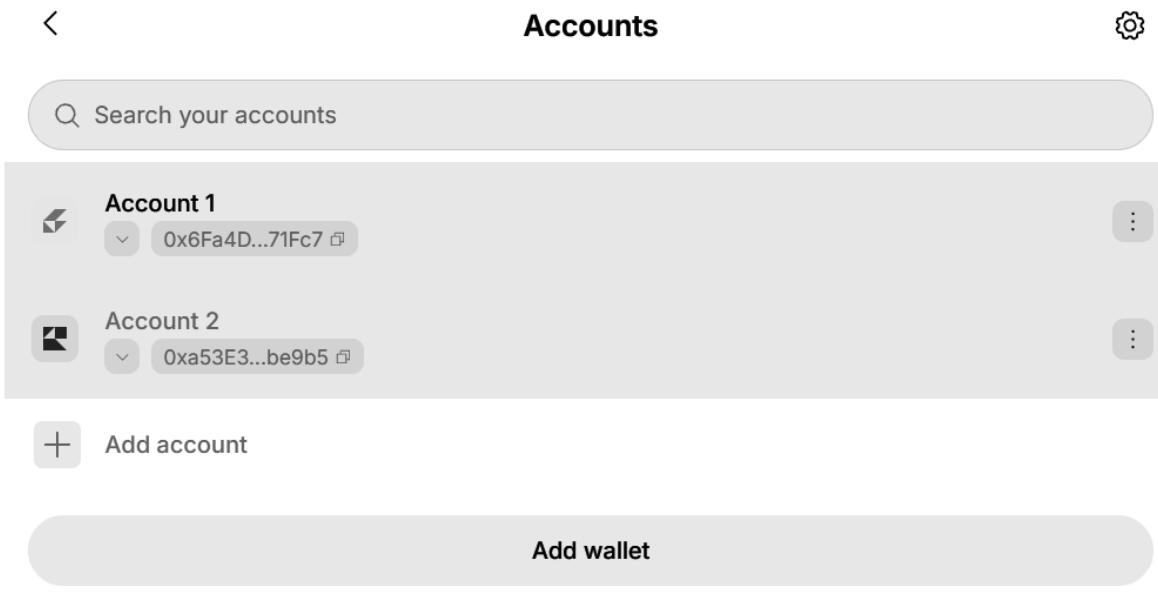
This experiment demonstrates the creation and configuration of accounts in MetaMask and the process of transferring cryptocurrency between two blockchain accounts. It provides practical knowledge of wallet addresses, transaction fees, transaction hashes, and blockchain transaction verification. It also illustrates how blockchain technology enables the secure and transparent recording of digital asset transfers.

Precautions

1. Use the official MetaMask website to download the wallet.
2. Keep the Secret Recovery Phrase and private keys confidential.
3. Verify the receiver's wallet address before confirming the transaction.
4. Ensure that the correct blockchain network is selected.
5. Check the transaction amount and gas fee before submitting.
6. Use test cryptocurrency for practice to avoid losing real funds.
7. Verify the transaction status using a blockchain explorer.
8. Never share wallet credentials with unauthorized persons.




MetaMask
Account 1
0x6Fa4D...71Fc7
0.0890 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Tokens
DeFi
NFTs
Network: Sepolia
SepoliaETH
-
—
0.0890 SepoliaETH
CLG
-
—
<0.00001 CLG
Home
Perps
Activity


Accounts
Search your accounts
Account 1
0x6Fa4D...71Fc7
Account 2
0xa53E3...be9b5
Add account
Add wallet

Sending

0.01 SepoliaETH

From
Account 1

To
Account 2

Network — Sepolia

Network fee — 0.0003 SepoliaETH

Speed — Market ~12 sec

Cancel | Confirm

Sent SepoliaETH

-0.01 SepoliaETH

Status	Confirmed
Date	Oct 21, 2026, 12:46 PM
From	Account 1 (0x6fa4d...71fc7)
To	Account 2 (0xa53e3...be9b5)
Network	Sepolia
Transaction ID	0x4f8b8...cdb52
Total amount	0.01 SepoliaETH

View on block explorer


MetaMask
Account 2
0xa53E3...be9b5
0.0100 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Tokens
DeFi
NFTs
Network: Sepolia
SepoliaETH
-
0.0100 SepoliaETH
Home
Perps
Activity